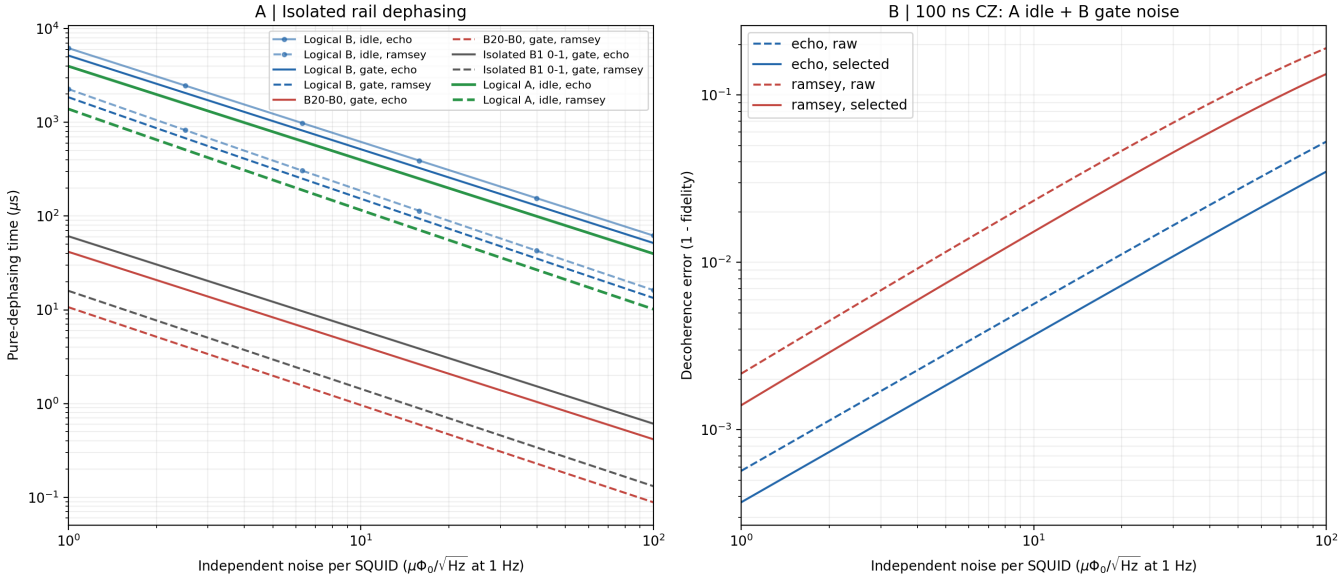
# 1 | Isolated rails and CZ decoherence error




Panel A: logical A at idle; logical B at idle/gate; B20-B0 and isolated B1 only at gate. Panel B: log-scale error for the ideal 100 ns B20-inclusive CZ, with independent A-idle and B-gate noise. The auxiliary has A in its pair ground. No coupler noise is added.

| Noise per loop | Echo raw | Echo selected | Ramsey raw | Ramsey selected |
|---|---|---|---|---|
| 1 | 99.9432% | 99.9631% | 99.7835% | 99.8599% |
| 10 | 99.4358% | 99.6330% | 97.6620% | 98.4734% |
| 100 | 94.7329% | 96.5134% | 80.9276% | 86.6408% |

First-order echo/Ramsey references substituted into exponential pure dephasing. Independent A-idle and B-gate generators added; correlations among logical and auxiliary states retained. B20 has A in pair ground. Equal noise amplitude per physical SQUID. No coupler noise, T1, quadratic/transverse noise or ramps in the central gate estimate.

# 2 | Isolated B-rail details

| SQUID parameter | B1 | B2 |
|---|---|---|
| Loaded EC/h (MHz) | 160.00 | 260.00 |
| Large junction EJ/h (GHz) | 22.04 | 14.12 |
| Small junction EJ/h (GHz) | 5.75 | 3.61 |
| Asymmetry | 0.5861 | 0.5932 |

| Position | Coherence | Echo (us) | Ramsey (us) |
|---|---|---|---|
| idle | Logical B | 619 | 186 |
| idle | B20-B0 | 5.09 | 1.19 |
| idle | Isolated B1 0-1 | 7.32 | 1.73 |
| gate | Logical B | 516 | 153 |
| gate | B20-B0 | 4.16 | 0.962 |
| gate | Isolated B1 0-1 | 6.09 | 1.43 |

Times use the sampled amplitude nearest10 microPhi0/sqrt(Hz) at1Hz. The isolated B1 reference removes B2 and C at the same B1 frequency and loaded EC. B20-B0 is the auxiliary phase coherence across the B component of the selected logical10/B20 transition, not a population lifetime.

Both bare B frequencies are calibrated to resonance independently at idle and gate. Differential detuning is first-order protected, but flux dependence of charge matrix elements changes the pair coupling. This calculation includes first-order longitudinal sensitivity only; quadratic and transverse noise are excluded.

# 3 | A-rail tunability comparison (alternative only)

The nominal A1/A2 design remains 4.00-5.00 GHz (1 GHz span), as used throughout this report. The alternative is 4.36-4.86 GHz (500 MHz total span), centered at the same 4.61 GHz idle setting. This comparison both narrows the range and recenters it. Loaded ECs, idle position, B design and the 100 ns exponential gate model are unchanged. No coupler noise is added.

| A transmon | Large EJ/h: nominal / alternative (GHz) | Small EJ/h: nominal / alternative (GHz) |
|---|---|---|
| A1 | 16.72 / 17.37 | 3.53 / 1.81 |
| A2 | 11.48 / 11.91 | 2.37 / 1.21 |

Isolated logical A at idle: dephasing times in microseconds. Noise amplitude is in $\mu\Phi_0/\sqrt{\text{Hz}}$ at 1 Hz per independent physical loop.

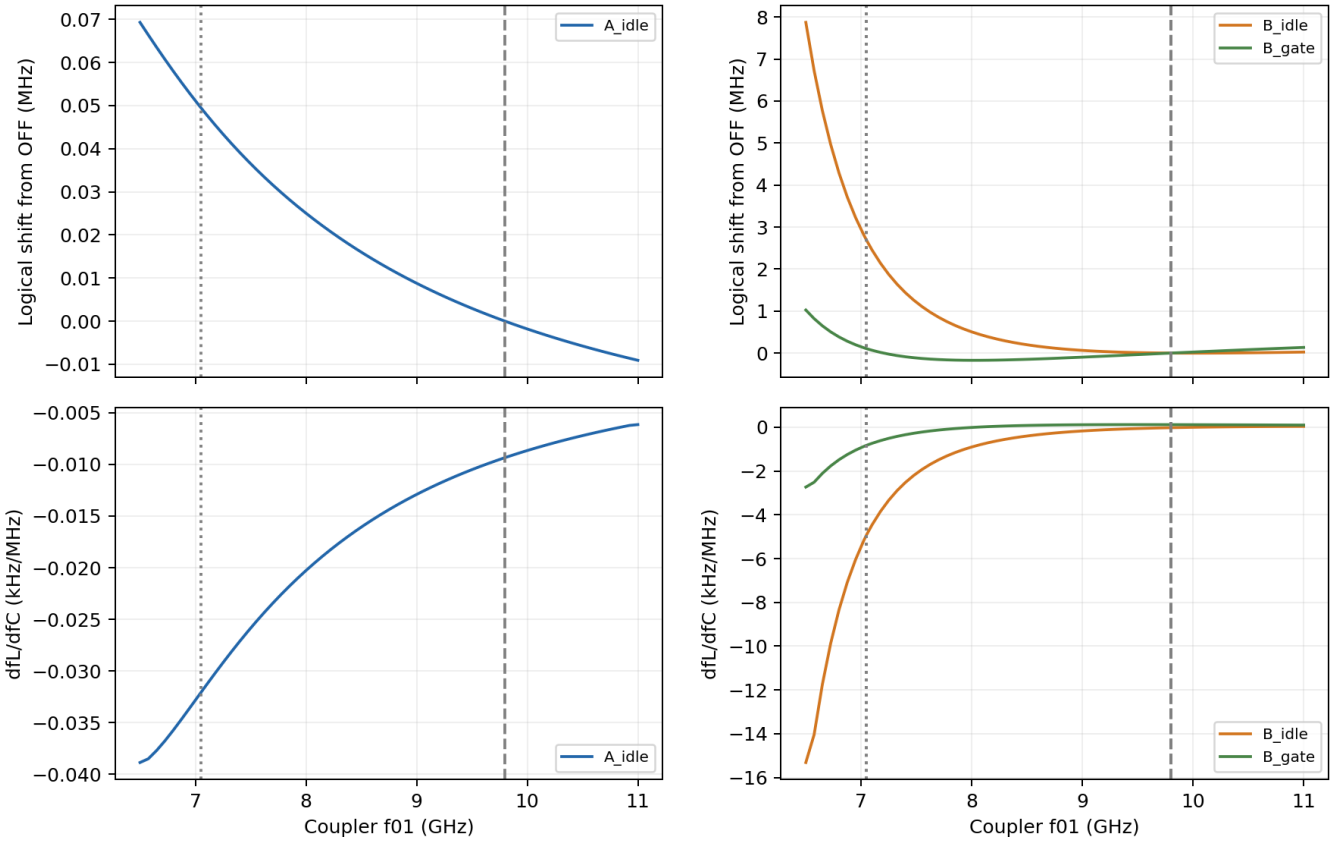
| Noise | Echo: 1 GHz | Echo: 500 MHz | Ramsey: 1 GHz | Ramsey: 500 MHz |
|---|---|---|---|---|
| 1 | 3969.08 | 7476.62 | 1388.42 | 2793.45 |
| 10 | 396.91 | 747.66 | 115.76 | 227.71 |
| 100 | 39.69 | 74.77 | 10.16 | 19.77 |

Full B20-inclusive CZ with A-idle and B-gate noise: fidelities below are percentages, corresponding to error 1-F in page 1 panel B.

| Noise / rates | Raw: 1 GHz | Raw: 500 MHz | Selected: 1 GHz | Selected: 500 MHz |
|---|---|---|---|---|
| 1 / echo | 99.9432% | 99.9499% | 99.9631% | 99.9675% |
| 1 / ramsey | 99.7835% | 99.8087% | 99.8599% | 99.8763% |
| 10 / echo | 99.4358% | 99.5017% | 99.6330% | 99.6764% |
| 10 / ramsey | 97.6620% | 97.9272% | 98.4734% | 98.6493% |
| 100 / echo | 94.7329% | 95.3132% | 96.5134% | 96.9087% |
| 100 / ramsey | 80.9276% | 82.5976% | 86.6408% | 87.9198% |

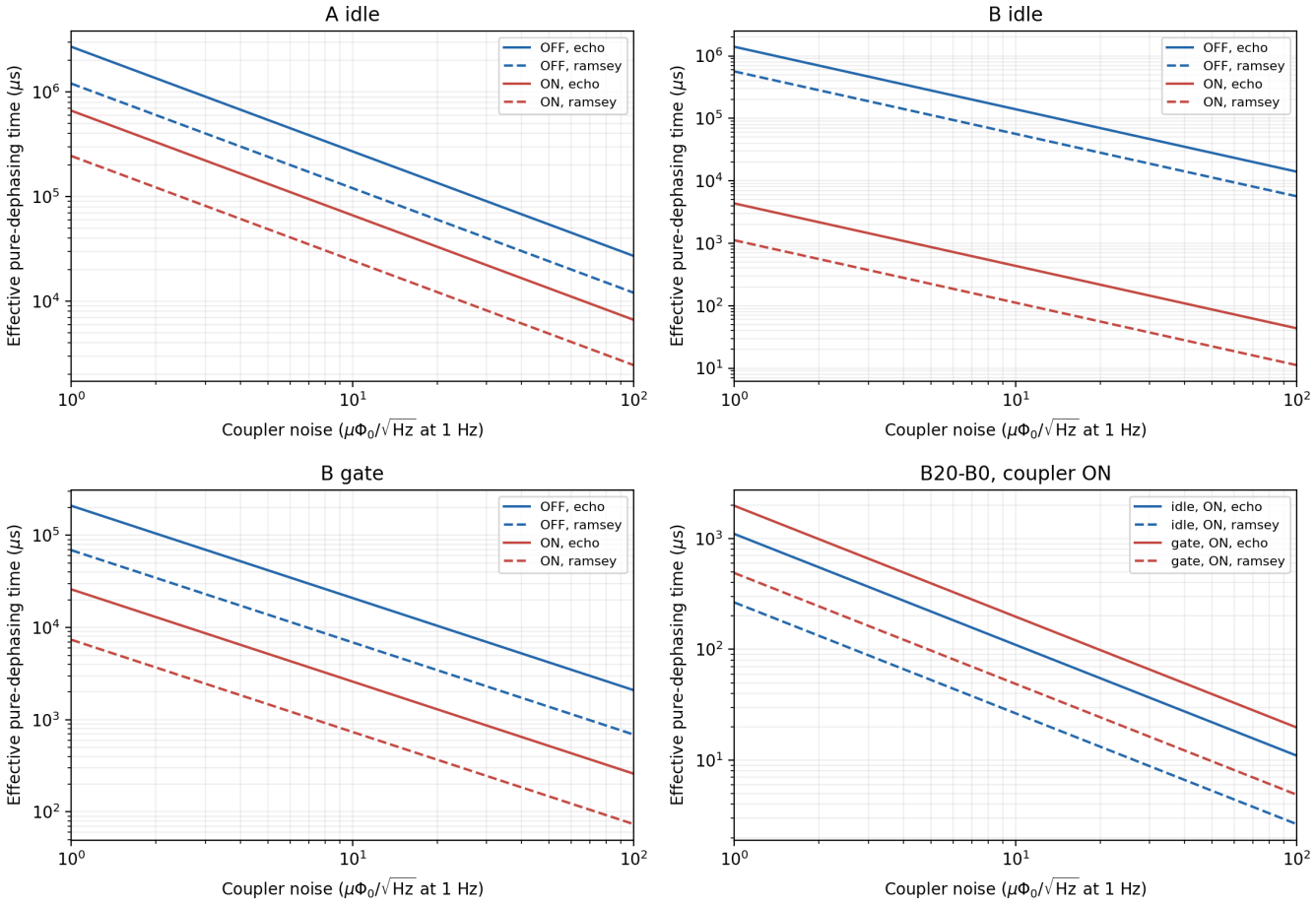
The narrower-range alternative approximately doubles logical A coherence. At noise amplitude 10, total raw and selected gate errors decrease by about 12%; B noise still contributes most of the error. This alternative is not adopted as the nominal design. Complete comparison values are retained in tuning-comparison.json.

# 4 | Separate coupler-induced dephasing




Rail EJs are noiseless. Differential rail bias is calibrated once at idle with C OFF, then held fixed for the common move to gate. Only C EJ changes along the sweep. Dashed/dotted lines denote OFF/ON. This differs from independently resonating the isolated B pair at every operating point; the comparison is not an identical-control subtraction.

# 5 | Coupler-induced dephasing times




Times are in microseconds. Reference amplitude100 microPhi0/sqrt(Hz) at1Hz is scaled inversely with amplitude for the adopted exponential-rate convention. The Ramsey cutoff logarithm is held at its reference value in this separate C-noise figure; isolated B Ramsey times are solved independently at every amplitude. B20 is shown only relative to B0 and only with C ON. No intrinsic rail noise or T1 is added here.

# 6 | Relative size, interpretation and checks

| Gate coherence | Rail Tphi (us) | C Tphi (us) | Rail rate / C rate |
|---|---|---|---|
| Logical B | 51.61 | 259.58 | 5.03 |
| B20-B0 | 0.42 | 19.74 | 47.49 |

Compare the listed rate ratios rather than assume a uniform order-of-magnitude reduction. Static dressing and different calibration conventions can matter separately from adding a noise source. Combined-noise decoherence and fidelity are intentionally omitted from this report.

Gate rates are phenomenological exponential substitutions of echo/Ramsey first-order 1/f reference times. Independent A/B generators add, retaining state correlations and the A-ground/B20 auxiliary. Rate matrices are checked for complete positivity. No T1, thermal population, gate-matrix-element noise, pulse ramps or baseline coherent error is included. Separate coupler-noise curves remain diagnostics, not an added gate-error source here. All inputs and derivatives remain in results.json.